Closed-Loop Graph Learning for Adaptive Sensor Selection in DOA Tracking

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***Abstract*—Efficient sensor selection is critical for resource-constrained target tracking, yet traditional heuristic methods often fail to capture the complex coupling between nonlinear filtering and long-term dynamics. We propose closed-Loop graph learning (CGL), a differentiable framework for adaptive sensor selection in direction-of-arrival (DOA) tracking. Specifically, a graph attention network models the spatial sensor topology while incorporating real-time tracking uncertainty as a closed-loop feedback signal. Under a fixed cardinality constraint, a differentiable selection layer generates sparse activation policies integrated into an extended kalman filter (EKF). The framework is optimized via a hybrid loss function that combines an information-utility and a closed-loop tracking loss. Simulations demonstrate that CGL achieves a superior accuracy-efficiency trade-off, providing near-optimal precision with a 12-fold reduction in overhead compared to optimization-based benchmarks.**

***Index Terms*—sensor selection, closed-Loop graph learning, graph attention networks, extended kalman filter.**

I. Introduction

In the landscape of distributed sensor networks [1], achieving precise target localization and tracking remains a paramount objective, particularly when operating under stringent resource limitations. The inherent scarcity of energy reserves, restricted communication throughput, and modest onboard processing power across sensor nodes render the simultaneous activation of the entire network impractical and inefficient [2]. Consequently, the challenge lies in devising a node selection mechanism that strategically identifies a sparse yet highly informative subset of sensors to maintain superior tracking fidelity without excessive resource expenditure.

Current research into sensor selection has branched into several established paradigms, ranging from decentralized heuristics to rigorous mathematical optimizations. In the adaptive node selection (ANS) approach [3], each node determines its activation status by contrasting its individual information gain against that of adjacent sensors, balancing energy consumption with tracking precision. Information-guided clustering groups sensors into subsets according to their collective information contribution and activates the most representative nodes per cluster, thereby optimizing resource allocation across multi-target tracking scenarios [4]. While such methods minimize the communication burden, they frequently overlook the global network topology, potentially resulting in suboptimal geometric configurations—often referred to as geometric dilution of precision (GDOP) [5]—which compromises tracking stability. To refine the geometric synergy between sensors, researchers have proposed the global node selection (GNS) [6] to assess the collective contribution of all candidate nodes. To address the combinatorial nature of the sensor selection problem, semidefinite programming (SDP) has been introduced as an effective relaxation framework, where binary selection constraints are relaxed into a convex domain and the Cramér–Rao Lower Bound (CRLB) is employed as the optimization criterion [7], [8]. However, the substantial computational latency associated with these solvers poses a significant barrier for real-time applications involving highly dynamic targets, where the recursive state estimation is typically maintained by an extended Kalman filter (EKF) [9].

With the development of deep learning theory [10], learning-based approaches have emerged as promising tools for efficiently approximating combinatorial sensor selection policies. Specifically, in [11], a deep neural network-based dynamic sensor selection method is proposed to learn input-dependent sensor subsets in an end-to-end manner, improving energy efficiency in wireless sensor networks while maintaining competitive performance in electroencephalography-based applications. However, learning-based sensor selection methods for DOA tracking, particularly those explicitly accounting for the recursive uncertainty propagation in nonlinear filters such as the EKF, remain scarce. To bridge the gap between the accuracy of optimization-based methods and the efficiency of learning-based approaches, we propose a closed-loop graph learning (CGL) framework that tightly integrates sensor selection with EKF-based tracking. By leveraging graph attention networks and explicitly feeding back EKF uncertainty into the selection policy, the proposed method enables end-to-end differentiable optimization of joint sensor selection and target tracking.

II. Measurement Model and Problem Formulation

A. Signal Model

Consider a two-dimensional sensor network consisting of N spatially distributed passive sensors with known positions $\boldsymbol{s}_i = [x_i, y_i]^\top, i = 1, \cdots, N$. A single moving target is characterized by a constant-velocity state vector

$$\mathbf{x}_k = [x_k, y_k, v_{x,k}, v_{y,k}]^\top \tag{1}$$

at discrete time index k, where (x_k, y_k) denotes the target position and $(v_{x,k}, v_{y,k})$ represents its velocity components.

The target dynamics follow a standard nearly constant velocity model

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{G}\mathbf{w}_k, \tag{2}$$

where $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \sigma_a^2 \mathbf{I}_2)$ represents zero-mean white Gaussian acceleration noise, and the state transition matrices are given by

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \frac{1}{2}\Delta t^2 & 0 \\ 0 & \frac{1}{2}\Delta t^2 \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix}, \tag{3}$$

resulting in a process noise covariance $\mathbf{Q} = \mathbf{G}(\sigma_a^2 \mathbf{I}_2)\mathbf{G}^\top$.

B. Retarded DOA Measurement Model

Each sensor provides a direction-of-arrival (DOA) measurement. Following the retarded DOA model [6], the effective DOA measurement at sensor i is:

$$z_{i,k} = \theta_{i,k} + \delta_{i,k} + n_{i,k}, \tag{4}$$

where $\theta_{i,k}$ is the geometric bearing angle from sensor i to the target position (x_k, y_k), $n_{i,k} \sim \mathcal{N}(0, \sigma_\theta^2)$ is the measurement noise. The motion-induced angular bias $\delta_{i,k}$, arising from the delay between signal emission and reception, is modeled as $\delta_{i,k} = \arcsin\left(\frac{v_k}{c}\sin(\theta_{i,k} - \psi_k)\right)$, where v_k and ψ_k are the target's instantaneous speed and heading. For a selected active sensor set $\mathcal{S}_k \subseteq \{1, \ldots, N\}$, the stacked observation model is defined as $\mathbf{z}_k = \mathbf{h}_{\mathcal{S}_k}(\mathbf{x}_k) + \mathbf{n}_k$.

C. Problem Formulation

The objective is to learn an optimal dynamic sensor selection policy π that selects a fixed subset of N_d sensors at each time step, expressed as:

$$\min_{\pi} \mathbb{E}\left[\sum_{k=1}^{T} \mathcal{L}(\hat{\mathbf{x}}_k, \mathbf{x}_k, \mathbf{m}_k)\right], \quad \text{s.t. } |\mathbf{m}_k|_0 = N_d, \forall k, \tag{5}$$

where T represents the time step of the trajectory, $\hat{\mathbf{x}}_k$ is the EKF state estimate conditioned on the history of selected measurements, $\mathbf{m}_k \in \{0,1\}^N$ is the activation mask, only measurements from the sensors where $m_{i,k} = 1$ are utilized in the EKF measurement update. $\mathcal{L}(\cdot)$ is a loss function of the proposed network. We propose a learning-based framework that enables end-to-end policy optimization within the recursive filtering process. The proposed architecture is detailed in the following section.

III. The Proposed Method

Fig. 1 illustrates the proposed closed-loop graph learning (CGL) framework, which establishes a tight coupling between the tracking state and the sensor selection policy. Specifically, a graph attention network (GAT) [12], [13] processes the spatial topology to produce a scoring vector $\mathbf{p}_k$. A differentiable straight-through estimator (STE) [14] then generates a binary mask $\mathbf{m}_k$, selecting N_d sensors. The EKF measurement update is executed using only these active sensors, and the resulting posterior state is fed back into the next iteration.

A. Graph Construction and Node Features

We model the sensor network as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with N nodes. To capture the geometric relationship between the target and sensors, we define a 3-dimension node feature vector for each sensor i at time k:

$$\mathbf{f}_{i,k} = \left[\tilde{d}_{i,k}, \cos(\theta_{i,k}), \sin(\theta_{i,k})\right]^\top, \tag{6}$$

where $\tilde{d}_{i,k}$ is the normalized distance between the predicted target position and sensor i, and $\theta_{i,k}$ is the relative bearing angle. To incorporate the sensor network's topology, we define edge attributes $\mathbf{e}_{ij} = (\mathbf{s}_i - \mathbf{s}_j)/L$, representing the normalized displacement between sensors i and j, where $\mathbf{s}_i = [x_i, y_i]^\top$ denotes the position of sensor i and L is a normalization constant chosen as the maximum inter-sensor distance in the network.

B. Backbone of the CGL

The node selection network is designed to fuse local spatial topology, global geometric context, and real-time tracking uncertainty. Initially, edge attributes are processed through a linear embedding layer to generate high-dimensional edge features $\mathbf{e}'_{ij}$. Local spatial features for each sensor are then extracted using a GAT layer, which computes multi-head attention over the local neighborhood $\mathcal{N}_i$:

$$\mathbf{h}_{i,k} = \mathrm{GAT}\big(\mathbf{f}_{i,k}, \mathbf{e}'_{ij} \mid j \in \mathcal{N}_i\big), \tag{7}$$

where $\mathbf{h}_{i,k}$ captures the local geometric importance of sensor i. Then, an attention-based global pooling mechanism computes a weighted context vector:

$$\mathbf{g}_k = \sum_{i=1}^{N} \alpha_{i,k}\mathbf{h}_{i,k}, \tag{8}$$

where $\alpha_{i,k} = \mathrm{softmax}(\mathrm{MLP}_{\mathrm{attn}}(\mathbf{h}_{i,k}))$ represents the adaptive contribution of each sensor to the global tracking state. Furthermore, we encode the log-normalized trace of the predicted covariance matrix to provide a scale-invariant representation of the tracking confidence, expressed as:

$$u_k = \mathrm{MLP}_{\mathrm{unc}}(\log(\mathrm{tr}(\mathbf{P}_{k|k-1}))). \tag{9}$$

where $\mathrm{MLP}_{\mathrm{unc}}$ a multilayer perceptron (MLP). The final selection score for sensor i is produced by a MLP that concatenates the local embedding, global context, and uncertainty feature:

$$p_{i,k} = \mathrm{MLP}_{\mathrm{score}}([\mathbf{h}_{i,k}, \mathbf{g}_k, u_k]). \tag{10}$$

To convert the continuous score vector $\mathbf{p}_k = [p_{1,k}, \ldots, p_{N,k}]^\top$ into a discrete decision, a selection operator identifies the N_d highest-scoring sensors. The operation be defined as:

$$\mathbf{m}_k = \text{Top-}N_d(\mathbf{p}_k), \tag{11}$$

where $m_{i,k} = 1$ if $p_{i,k}$ is among the N_d largest elements, and $m_{i,k} = 0$ otherwise. A STE is used during backpropagation to enable gradient-based learning. By utilizing the EKF's inherent recursive state $(\hat{\mathbf{x}}, \mathbf{P})$ as a sufficient statistic for the tracking process, this architecture effectively captures temporal dynamics without the need for additional recurrent units.

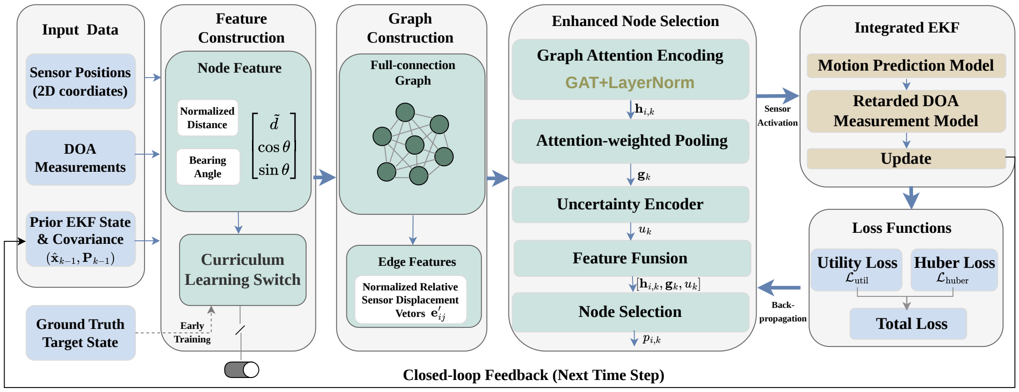

Fig. 1. The closed-loop graph learning framework for sensor selection in DOA tracking.

C. Differentiable EKF Layer

The tracking process is governed by a differentiable EKF layer, the prediction step yields the predicted state $\hat{\mathbf{x}}_{k|k-1}$ and covariance $\mathbf{P}_{k|k-1}$. Given the activation indices $\mathcal{S}_k = \{i \mid m_{i,k} = 1\}$, the innovation vector $\mathbf{y}_k$ is computed by:

$$\mathbf{y}_k = \mathbf{z}_{\mathcal{S}_k,k} - \mathbf{h}_{\mathcal{S}_k}(\hat{\mathbf{x}}_{k|k-1}), \tag{12}$$

where $\mathbf{z}_{\mathcal{S}_k,k}$ represents the DOA measurements from the N_d active sensors. The observation Jacobian $\mathbf{H}_k$ and the measurement noise matrix $\mathbf{R}_k$ are formed by indexing the full sensor parameters according to $\mathcal{S}_k$. The Kalman gain $\mathbf{K}_k$, which determines the optimal weight assigned to the new measurements, is then calculated as:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}_k^\top(\mathbf{H}_k\mathbf{P}_{k|k-1}\mathbf{H}_k^\top + \mathbf{R}_k)^{-1}. \tag{13}$$

Finally, the posterior state $\hat{\mathbf{x}}_{k|k}$ and covariance $\mathbf{P}_{k|k}$ are updated by incorporating $\mathbf{y}_k$:

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k\mathbf{y}_k, \tag{14}$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_{k|k-1}. \tag{15}$$

D. The Loss Function

The framework is optimized with a hybrid loss that combines a utility-based term and a supervised tracking term. The utility loss promotes informative sensor geometries, while the tracking loss anchors the policy to the closed-loop filtering objective, preventing convergence to GDOP-unaware solutions. For the utility loss, we employ the Fisher information matrix to guide the network toward geometrically favorable sensor constellations, formulated as:

$$\mathcal{L}_{\text{util}} = \frac{1}{T}\sum_{k=1}^{T}\left(\log\operatorname{tr}(\mathbf{J}_k^{\text{post}}) - \log\det(\mathbf{J}_k^{\text{post}})\right), \tag{16}$$

where $\mathbf{J}_k^{\text{post}} \triangleq \mathbf{P}_{k|k}^{-1}$ denotes the posterior information matrix under the Gaussian posterior assumption of the EKF.

To ensure robust trajectory tracking, we employ a supervised Huber loss on the position error:

$$\mathcal{L}_{\text{huber}} = \frac{1}{T}\sum_{k=1}^{T}\text{Huber}(\hat{\mathbf{x}}_k^{(1:2)} - \mathbf{x}_k^{(1:2)}, \delta), \tag{17}$$

where δ is a threshold that reduces the influence of outliers during the initial phases of training. The final loss is $\mathcal{L} = \lambda_u\mathcal{L}_{\text{util}} + \lambda_h\mathcal{L}_{\text{huber}}$, where λ_u and λ_h are weighting coefficients that balance the utility and tracking objectives.

IV. Simulation Results

A. Experimental Setup and Scenario Configuration

The proposed CGL framework is evaluated under the simulation setup detailed in Table I. Its performance is compared against three baseline selection strategies: GNS [6], ANS [3], and SDP [8].

TABLE I
SIMULATION PARAMETERS AND SCENARIO CONFIGURATION

Parameter type	Symbol / Item	Value / Description
Network layout	Surveillance area	$1000\,\text{m} \times 2000\,\text{m}$
	Total sensors, N	50 (uniformly distributed)
	Active nodes per step, N_d	4
Target motion	Sampling interval	1.0 s
	Initial position error	50 m (per axis)
	Initial velocity error	5 m/s (per axis)
	Initial covariance, $\mathbf{P}_0$	$\text{diag}(50^2, 50^2, 5^2, 5^2)$
Noise & physics	Process noise, σ_a	$0.5\,\text{m/s}^2$
	Measurement noise, σ_θ	5° (Gaussian)
	Speed of sound, c	347 m/s

B. Training Configurations

The training process is implemented using PyTorch Geometric and follows a structured curriculum to ensure filter stability. The key configurations of the neural network and the optimization process are listed in Table II.

Specifically, each data sample corresponds to an independent tracking episode comprising T consecutive time steps over a network of N spatially distributed sensors. The target state at time k, denoted by $\mathbf{x}_k$ as in (1), evolves according to the model in (2). The dataset is randomly partitioned into training and testing subsets with an 80%/20% ratio, ensuring that episodes from different trajectories remain separate.

TABLE II
NEURAL NETWORK ARCHITECTURE AND TRAINING STRATEGY

Category	Hyperparameter	Configuration
Optimization	Optimizer	Adam ($lr = 1e-4$)
	Epoch	50
	Loss Function	Utility Loss & Huber Loss
	Loss Trade-off	$\lambda_{\mathrm{u}} = 1$, $\lambda_{\mathrm{h}} = 0.5$
	Gradient Clipping	1.0
Strategy	Dataset Size	200 Trajectories $\times$ 100 Steps
	Teacher Forcing	First 10 Epochs (GT-guided)
	Closed-loop Phase	Epoch 11 to 50 (EKF-guided)

C. Dynamic Node Activation Analysis

In this experiment, we randomly selected a trajectory for testing to verify whether the proposed method can dynamically reconfigure the sensor subset in response to target movement. Snapshots of the tracking process are captured at four distinct intervals: $k = 2, 30, 65$ and 100. Each snapshot visualizes the historical ground-truth path, the EKF-estimated trajectory, the distribution of all available sensors, and the specific nodes activated by the proposed method at the current instant.

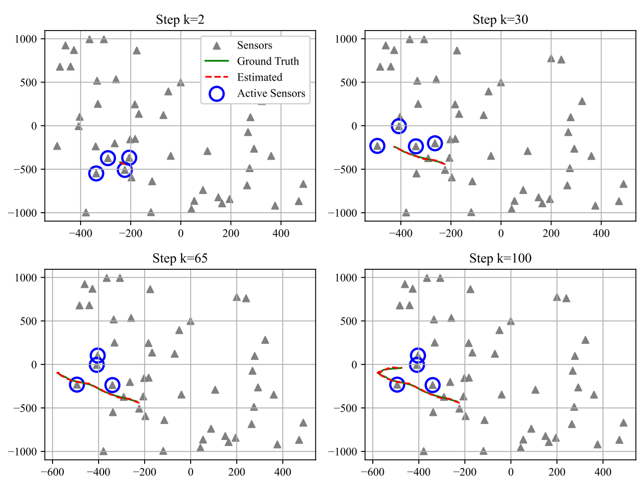

Fig. 2. The dynamic tracking performance and node activation under different snapshots.

Fig. 2 illustrates the dynamic tracking performance and node activation for a representative test trajectory. The proposed method demonstrates high-fidelity tracking, with the estimated path closely aligning with the ground truth despite the non-linearities of bearings-only measurements and retarded DOA effects. The GAT-based policy autonomously reconfigures the sensor subset in real-time, consistently prioritizing nodes that provide wide angular baselines relative to the target's predicted position. By incorporating the trace of the EKF covariance matrix $\mathbf{P}$ as a context feature, the model effectively balances proximity and orientation diversity and ensures filter stability throughout the tracking duration.

D. Performance Comparison

To evaluate the robustness of the proposed method, a Monte Carlo simulation is conducted over 50 independent test trajectories, each spanning 100 time steps.

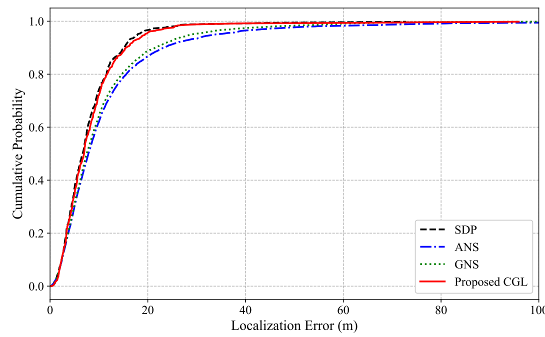

Fig. 3. The CDF of the localization errors under different algorithms.

The cumulative distribution function (CDF) of the localization errors for the evaluated methods is illustrated in Fig. 3. The proposed framework exhibits a significantly steeper ascent in the low-error region compared to GNS and ANS, indicating a superior concentration of tracking precision. The proposed method achieves a median error that closely approximates the performance of the SDP-based optimization, which serves as the theoretical performance bound. This demonstrates that the learning-based policy successfully identifies optimal sensor configurations across varied dynamic scenarios.

TABLE III
COMPARATIVE STATISTICAL PERFORMANCE AND COMPUTATIONAL EFFICIENCY

Method	RMSE (m)	Median Error (m)	Time per Step (ms)
GNS	8.22	7.38	360.05
ANS	9.14	7.87	21.91
SDP	7.55	5.55	1071.17
Proposed CGL	7.58	5.70	88.82

As shown in Table III, CGL performs nearly on par with SDP in both RMSE and Median Error, while achieving a 12-fold reduction in latency. Compared to GNS and ANS, CGL reduces RMSE by 7.8% and 17.1%, and improves Median Error by 22.8% over GNS. These results confirm that the hybrid loss and closed-loop feedback enable near-optimal selection without the prohibitive cost of iterative optimization.

CONCLUSION

In this paper, we presented the CGL method for adaptive sensor selection in DOA tracking. By integrating a GAT with a differentiable EKF layer, we achieved end-to-end optimization of selection policies directly within the filtering recursion. Experimental results demonstrate that CGL achieves a near-optimal accuracy-efficiency trade-off. Furthermore, by utilizing the EKF's recursive state as a sufficient statistic, the framework captures long-term temporal dynamics without the need for additional recurrent units.

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